



The first mission to measure electrons, ions, and multi-thermal plasma simultaneously in solar flares with sub-second EUV imaging and high-sensitivity keV–MeV spectroscopy.

Payload

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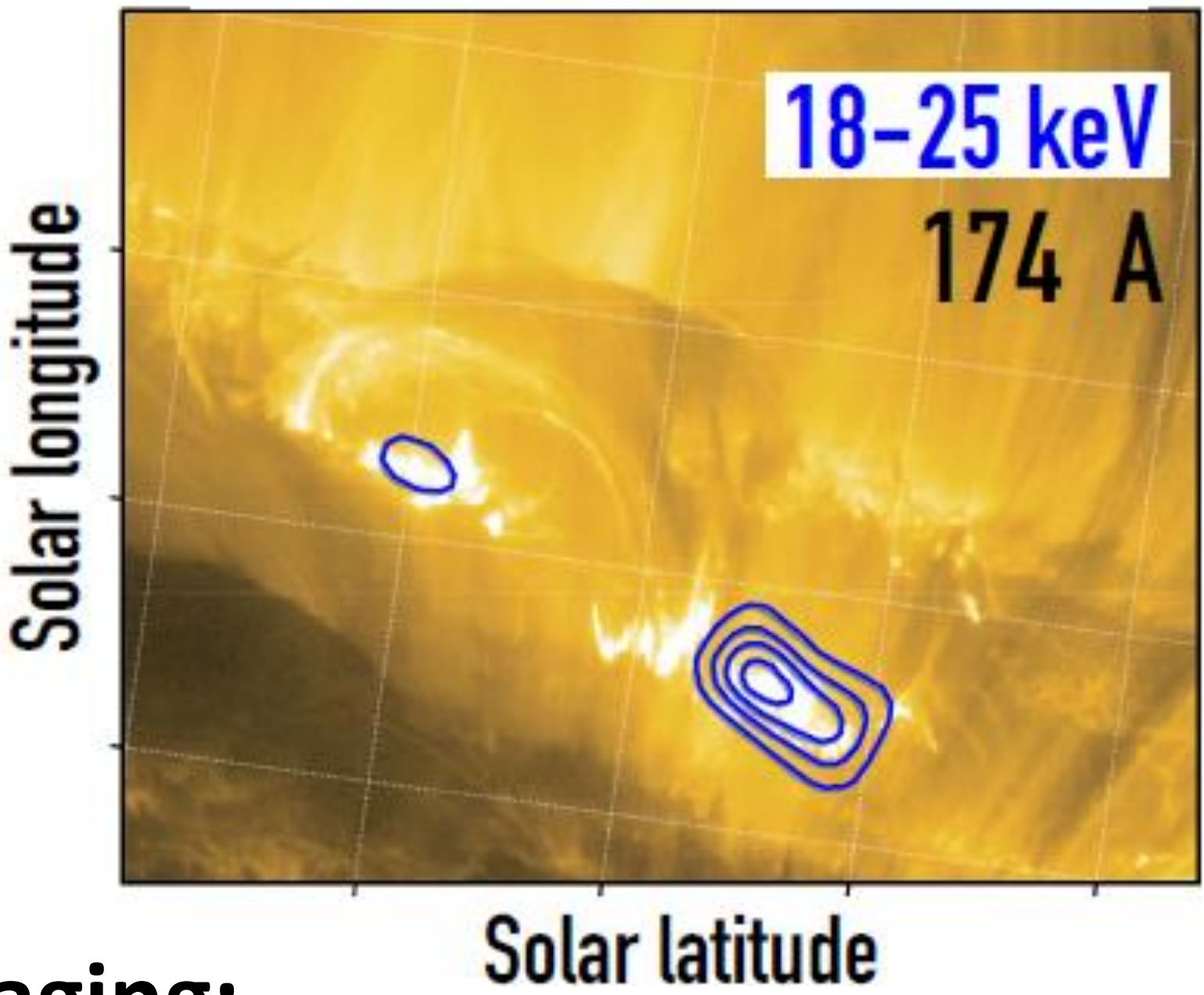
HiFI: High resolution Flare Imager

- EUV imaging capturing ribbons, kernels, nanojets and flare loops.

LASSAN: Large Area Spectrometer for Solar Accelerated Nuclei

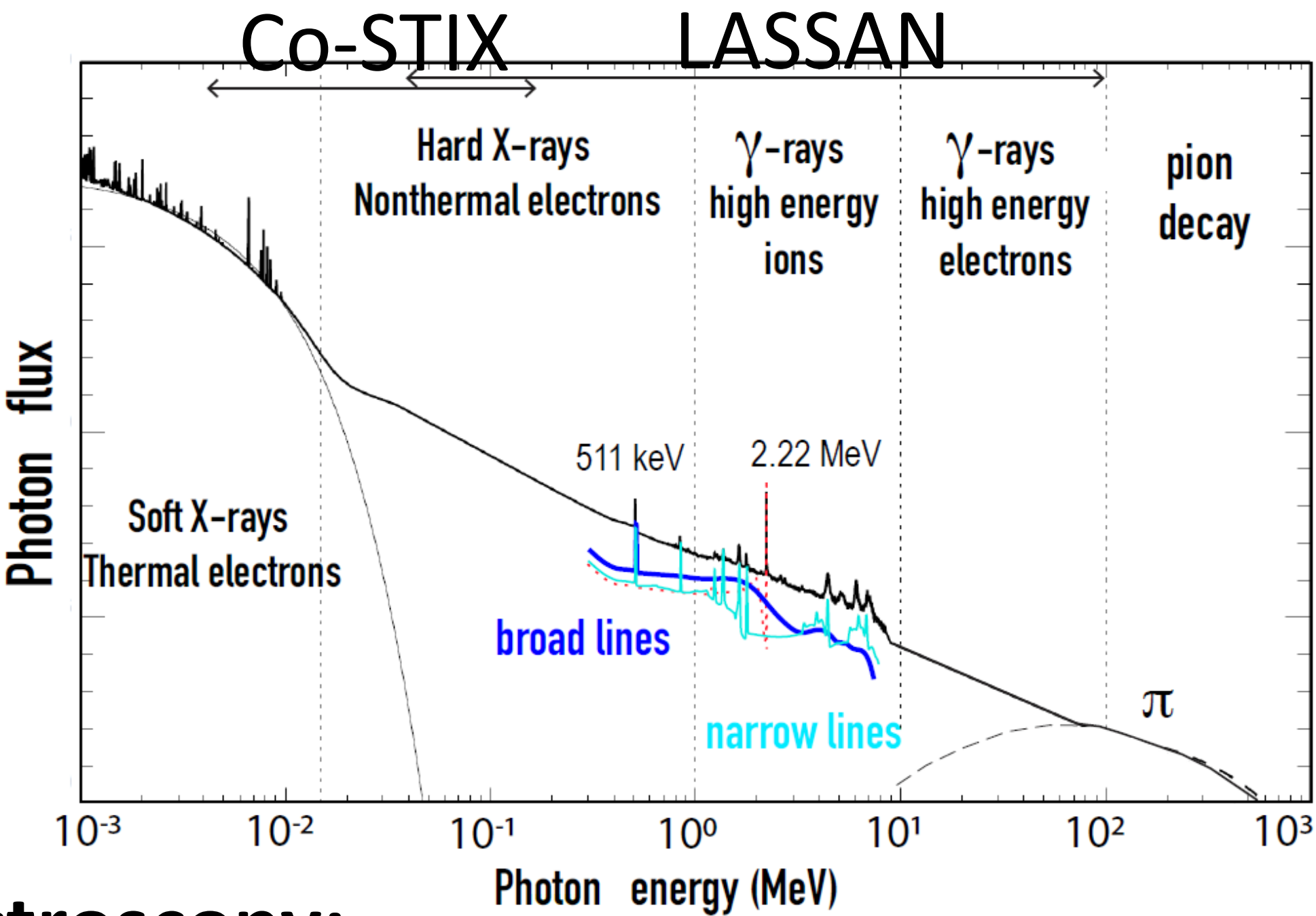
- keV–MeV spectroscopy, electrons & ions, pion continuum, γ -ray lines
- Co-STIX – context X-ray spectroscopic imager

Instrument Capabilities



Imaging:

- 304Å, 171Å, 131Å, 94Å → (0.1–10 MK)
- 0.14" px (≈100 km)
- 0.5s cadence
- Unsaturated EUV imaging
- Context X-ray imaging

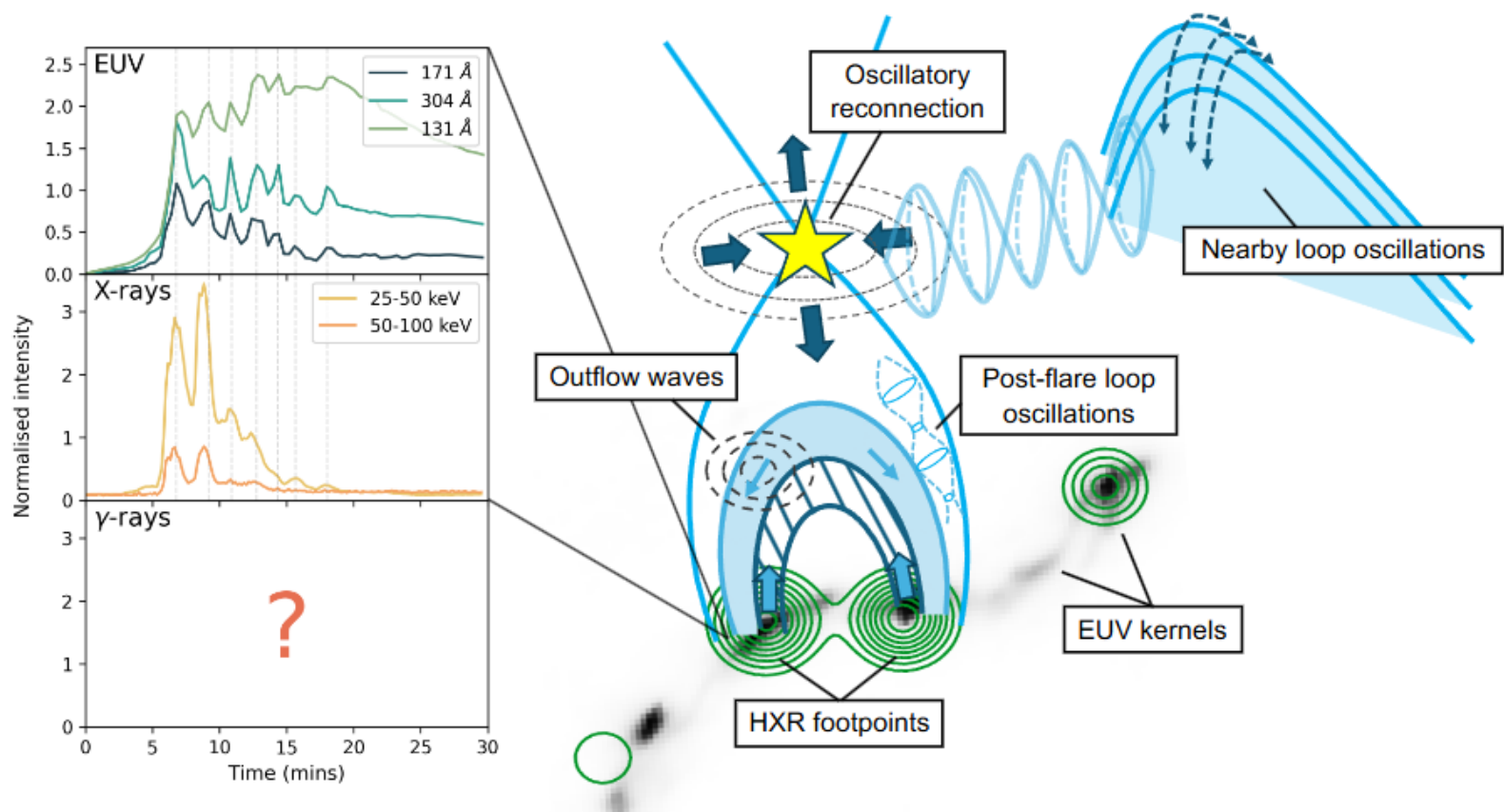
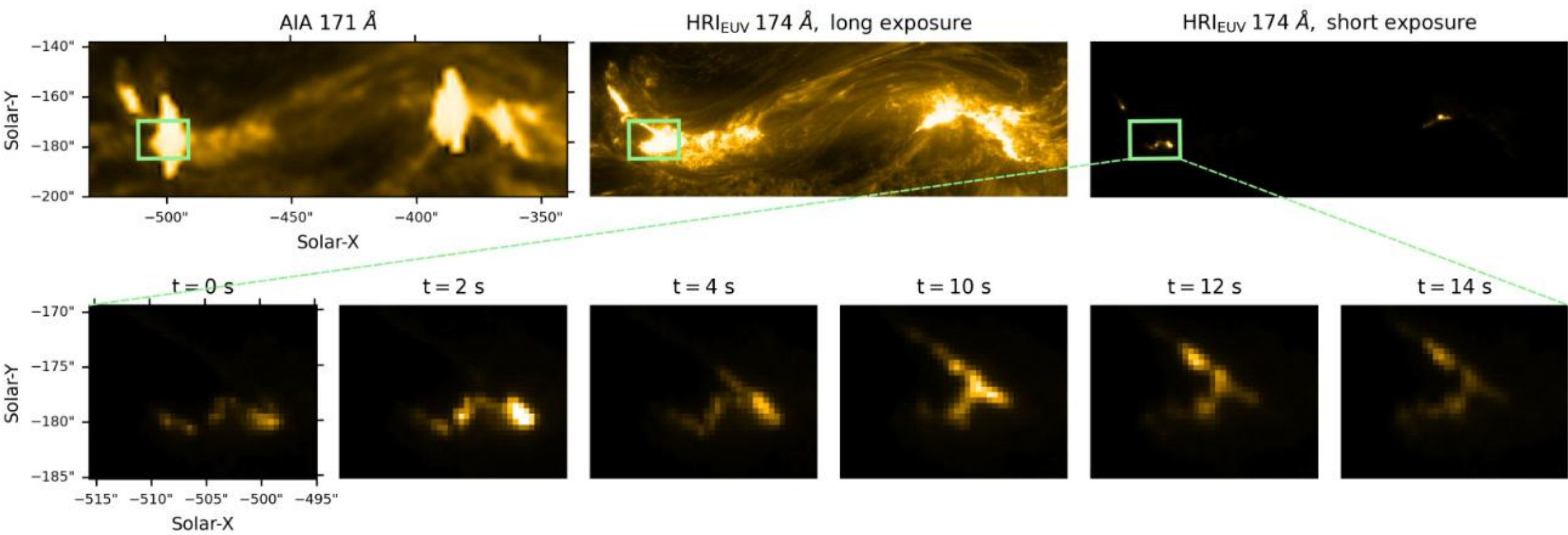


Spectroscopy:

- 4 keV – 100 MeV range
- 1.5% resolution at 6.1 MeV
- Large effective area (~1100 cm² at 2.2 MeV)
- Bremsstrahlung, γ -ray lines, pion continuum, HXR spectral evolution

Measurement Breakthroughs

- Rapid dynamics and sub-structure within solar flare ribbons, observable with saturation-free, short-exposure imagery .
- Evolution of ion spectra during flares and sub-second change in electron spectra and thermal plasma



Science Questions

1 How are electrons and ions accelerated during solar flares?

- LASSAN's broad keV-MeV coverage = simultaneous ion/electron diagnostics

2 How is solar flare energy transported and dissipated ?

- HiFI resolves kernels, ribbons, evaporation, conduction/waves at <1 s

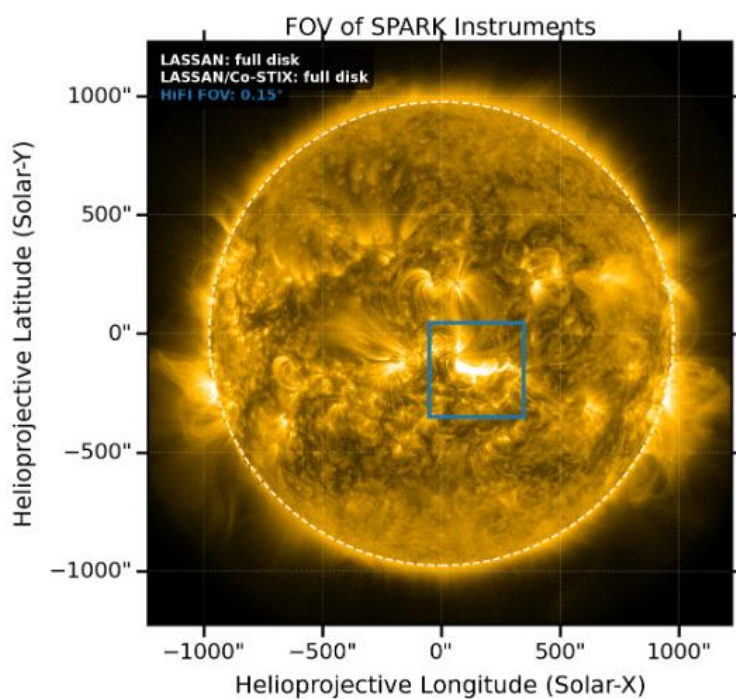
3 How do magnetic reconnection, waves, and plasma instabilities interact during flare energy release?

- Sub-second EUV + HXR timing = QPP regime unlocked

Mission Profile

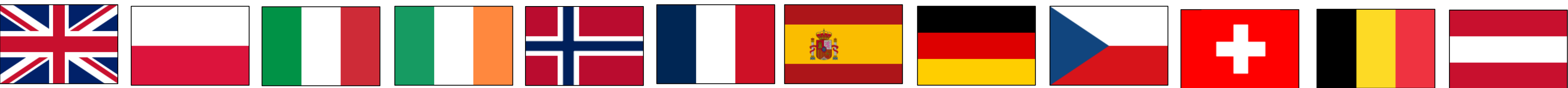
- Launch: Ariane 62, 2041
- Orbit: Halo at L1
- Mission lifetime: 4 + 3 years
- Payload: LASSAN + HiFI
- Telemetry: ~1000 Gbit/day
- Pointing stability: <0.1" over 1 s

Field of View



Why SPARK is a Step-Change

- First routine ion + electron acceleration flare measurements.
- First unsaturated, sub-second, multi-thermal EUV imaging of flare kernels, ribbons, nanojets.
- First mission to link MeV ions, keV electrons, and EUV plasma in real time.
- Transforms understanding of plasmoids , QPPs, current-sheet fragmentation, conduction vs wave heating, and flare onset.



Mission proposal to the ESA M8 2025 call